

Altermagnets Enable Gate-Switchable Helical and Chiral Topological Transport with Spin-Valley-Momentum-Locked Dual Protection

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这篇论文讲述了一个关于**“电子高速公路”**的突破性发现。想象一下，我们正在设计一种全新的交通系统，让电子（就像汽车）在材料中跑得更快、更稳，而且还能随意改变行驶规则。

这项研究的核心主角是一种叫做**“交替磁体”（Altermagnets）**的神奇新材料。为了让你更容易理解，我们可以把这篇论文的内容拆解成几个生动的比喻：

1. 以前的难题：脆弱的“双车道”

在传统的量子材料中，科学家试图利用“自旋霍尔效应”来制造一种特殊的**“双车道高速公路”**。

- **规则：**在这个公路上，电子必须遵守“左行右行”的规则（比如，自旋向上的车只能往左开，自旋向下的车只能往右开）。
- **问题：**这条公路非常脆弱。只要路上出现一点点“磁性路障”（磁性杂质），或者路面稍微有点颠簸，这种完美的交通规则就会崩溃，电子就会乱跑，导致交通堵塞（电阻变大），原本无损耗的传输就消失了。

2. 新发现：给电子加上“双重保险”

这篇论文提出，利用**“交替磁体”，我们可以给这条公路加上“双重保险”**。

- **什么是交替磁体？** 想象一个巨大的棋盘，黑格子和白格子上分别住着性格相反的“电子居民”。虽然它们整体看起来是平衡的（没有净磁性），但它们内部却有着强烈的“阵营”区分。
- **双重锁定 (SVML)：**在这篇论文设计的材料里，电子不仅被锁定了**“方向”（自旋），还被锁定了“阵营”**（谷/Valley）。
比喻：以前电子只是“左手车”或“右手车”。现在，电子变成了“左手红队车”和“右手蓝队车”。
效果：这种“自旋 - 谷 - 动量”的三重锁定，就像给电子穿上了防弹衣。即使路上有普通的障碍物（非磁性干扰）或者远处的磁性波动，电子依然能稳稳地沿着自己的车道行驶，不会撞车。这就是论文中提到的**“双重拓扑保护”**。

3. 魔法开关：一键切换“双车道”和“单行道”

这项研究最酷的地方在于，科学家发现了一个**“魔法开关”**（通过电压控制），可以随意改变这条公路的形态：

- **模式 A：双车道（螺旋态）** 当开关处于“关闭”状态时，公路是**双向**的。红队车往左，蓝队车往右。这就像传统的量子自旋霍尔效应，非常稳定，适合做无损耗传输。
- **模式 B：单行道（手性态）** 当我们给材料施加一个特定的**电压**（就像给路面加了一个斜坡），神奇的事情发生了：其中一个“阵营”的公路被强行填平了（变成了普通路面），只剩下另一个阵营的公路在运行。**结果：**双车道瞬间变成了**单向高速（单行道）**。所有的车都只朝一个方向跑，而且速度极快，完全不受任何干扰（包括那些以前能破坏双车道的磁性路障）。
反转：如果你把电压反过来，单行道的方向也会反过来（比如从只允许红队车跑，变成只允许蓝队车跑）。

4. 现实中的“魔法材料”

科学家不仅在理论上画出了蓝图，还通过超级计算机在现实世界中找到了具体的“建筑材料”：

- 他们发现像**单层 V₂TeO**（一种含有钒、硫、碲、氧的薄膜）这样的材料，天生就具备这种能力。
- 通过简单的“化学装修”（比如把其中一个原子换成钛原子），就可以人为地制造出那个“魔法开关”，实现从双车道到单行道的切换。

总结：这意味着什么？

这篇论文就像是在告诉未来的工程师：

“我们找到了一种**可编程的电子高速公路**。它平时是双向的，非常稳定；如果你需要它变成单向的超级高速公路（抗干扰能力更强），只要按下一个电压按钮就能瞬间切换。而且，这种切换不需要改变材料本身，也不需要极端的低温环境。”

这对未来的影响：

这为制造**超低功耗、抗干扰能力极强**的新一代电子芯片、量子计算机和自旋电子器件铺平了道路。它让“电子交通”变得更加智能、灵活和可靠，是连接基础物理和未来量子技术的一座重要桥梁。

这篇论文提出并建立了一个统一的、基于对称性的理论框架，利用**交变磁体（Altermagnets, AMs）独特的自旋劈裂特性与谷拓扑（Valley Topology）相结合，实现了在单一材料平台内通过电场调控，在螺旋（Helical）和手性（Chiral）**拓扑相之间进行可逆切换。

以下是该论文的详细技术总结：

1. 研究背景与核心问题 (Problem)

- 传统量子自旋霍尔效应 (QSH) 的局限性：**传统的 QSH 效应依赖时间反演对称性 (TRS) 保护，虽然具有无耗散的螺旋边缘态，但其自旋通道被自旋轨道耦合 (SOC) 纠缠，导致自旋霍尔电导无法严格量子化。此外，微弱的磁性杂质会破坏 TRS，导致边缘态不稳定，量子化输运被破坏。
- 传统量子反常霍尔效应 (QAH) 的局限：**将 QSH 转化为 QAH 通常需要引入长程铁磁序来打破 TRS，但这往往需要精细调节能隙重开，限制了材料实现、工作温度和器件的可调性。
- 挑战：**如何在一个材料平台中，既实现具有双重拓扑保护的鲁棒螺旋态，又能通过电控手段将其转换为抗所有类型无序的手性态，并实现自旋 - 谷极化的确定性选择。

2. 方法论 (Methodology)

- 理论框架：**结合交变磁体的动量空间交替自旋极化 (Alternating Spin Splitting) 与谷拓扑。利用交变磁体中由晶体旋转对称性 (CRS) 连接的谷 (α 和 β 谷)。
- 模型构建：**构建了一个基于二维正方晶格反铁磁 (AFM) 的紧束缚模型。引入自旋轨道耦合 (SOC) 打开拓扑能隙，形成**交变磁量子自旋谷霍尔 (AMQSVH) **相。引入**子晶格交错势 (Sublattice-staggered potential, U)**，该势场可通过外部电场 (栅压) 连续调控，用于打破晶体旋转对称性。
- 数值模拟与计算：**使用 Landauer-Büttiker 形式和格林函数方法进行大规模量子输运模拟，测试不同无序类型（非磁性/磁性，短程/长程）下的电导率。利用第一性原理计算 (DFT) 验证具体材料体系（如单层 $V_2\text{TeO}$ 和 VO 家族）的物理性质。

3. 关键贡献与机制 (Key Contributions & Mechanisms)

- 双重拓扑保护 (Dual Topological Protection)**：在 AMQSVH 相中，边缘态表现为**自旋 - 谷 - 动量锁定 (SVML)**。定义了一个复合自旋 - 谷陈数 $C_{\text{sv}} = 2$ (由自旋陈数 C_s 和谷陈数 C_v 共同决定)。这种双重锁定使得自旋混合被强烈抑制，边缘态对非磁性无序和长程磁性涨落具有极强的鲁棒性 (这是传统 QSH 所不具备的)。
- 电控相变机制 (Gate-Switchable Phase Transition)**：利用子晶格交错势 U 打破连接两个反向传播 SVML 模式的晶体旋转对称性。 **$U=0$** ：系统处于 AMQSVH 相，具有螺旋边缘态。 **$U \neq 0$ (正或负)**：选择性地使其中一个谷 (α 或 β) 的能隙拓扑平庸化 (Trivialize)，而保留另一个谷的拓扑非平庸性。**结果**：螺旋态转化为**交变磁量子反常霍尔 (AMQAVH) **相 ($C_{\text{sv}}=1$)，仅保留单一的手性边缘通道。
- 自旋 - 谷极化切换**：通过反转 U 的符号，可以切换传输的手性边缘态的自旋 - 谷极化方向 (例如从 α 谷的自旋向下切换到 β 谷的自旋向上)。

4. 主要结果 (Results)

- 输运鲁棒性验证**：**AMQSVH 相**：在模拟中显示，该相在非磁性无序 (短程/长程) 和长程磁性无序下保持完全量子的自旋电导。仅在短程磁性无序 (安德森型，能同时引起自旋翻转和谷间散射) 下量子化被破坏。**AMQAVH 相**：一旦进入手性相，系统对**所有类型**的无序 (包括短程磁性无序) 均表现出鲁棒的量子化输运。
- 材料实现**：第一性原理计算确认了单层 **$V_2\text{TeO}$** 和 **VO** 家族是理想的候选材料。在 $V_2\text{TeO}$ 中，SOC 打开约 20 meV 的能隙，形成 AMQSVH 相。通过子晶格掺杂 (例如用 Ti 替换 V_A 或 V_B 位点的 V 原子) 引入等效的交错势 U ，成功实现了向 AMQAVH 相的转换，并观察到单一的手性边缘态。

5. 科学意义 (Significance)

- 新型拓扑器件平台**：确立了交变磁体作为电荷、自旋和谷自由度上可电控编程的拓扑器件平台。
- 克服传统缺陷**：解决了传统 QSH 对磁性无序脆弱的问题，同时避免了传统 QAH 对长程铁磁序和精细能隙调节的依赖。
- 通用性**：该机制基于对称性和能带拓扑，而非特定化学性质，因此可扩展至光子学、声子学、磁子学和超导系统。
- 应用前景**：为开发低耗散、电可编程的拓扑电路和多功能量子器件 (如自旋 - 谷阀、拓扑晶体管) 提供了坚实的理论基础和材料方案。

总结：该论文通过理论创新和材料设计，展示了如何利用交变磁体的独特对称性，实现从受双重保护的螺旋态到全鲁棒手性态的电控切换，为下一代自旋电子学和谷电子学器件开辟了新途径。

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类似论文

Probing Boron Vacancy Defects in hBN via Single Spin Relaxometry

该研究通过将金刚石中的单氮 - 空位 (NV) 色心与扫描探针显微镜相结合，利用交叉弛豫机制间接探测六方氮化硼 (hBN) 中硼空位 ($\text{B}^{\cdot-}$) 缺陷的电子自旋共振，从而实现了无需光学激发的纳米级缺陷密度定量成像及超精细结构分辨。

Alex L. Melendez, Ruotian Gong, Guanghui He + 14 more2026-03-06  cond-mat.mes-hall

Evidence of Ultrashort Orbital Transport in Heavy Metals Revealed by Terahertz Emission Spectroscopy

该研究利用飞秒太赫兹发射光谱和楔形薄膜技术，首次直接证实了钨、钽、铂等重金属中存在比自旋更短的超短轨道平均自由程，并确认了体相逆轨道霍尔效应是主导机制，从而解决了轨道电子学领域的核心争议。

Tongyang Guan, Jiahao Liu, Wentao Qin + 4 more2026-03-06  cond-mat.mes-hall

Ordering the topological order in the fractional quantum Hall effect

本文以拓扑序理论中的全局对称性及其反常（特别是一形式对称性）为核心组织原则，根据霍尔电导值约束低能有效理论，从而唯一地确定了绝大多数实验已发现的分数量子霍尔效应系统的最小拓扑序。

Meng Cheng, Seth Musser, Amir Raz + 2 more2026-03-06  cond-mat.mes-hall

HSG-12M: A Large-Scale Benchmark of Spatial Multigraphs from the Energy Spectra of Non-Hermitian Crystals

本文提出了名为 Poly2Graph 的高效开源工具，构建了包含 1700 万张空间多重图的 HSG-12M 数据集，首次将非厄米晶体能谱自动转化为几何图数据，填补了现有图基准缺乏空间几何信息的空白，并为凝聚态物理的数据驱动发现及几何感知图学习开辟了新途径。

Xianquan Yan, Hakan Akgün, Kenji Kawaguchi + 2 more2026-03-06  cond-mat.mes-hall

Laser-induced topological phases in monolayer amorphous carbon

该研究提出利用圆偏振激光驱动单层非晶碳进入非平衡态，成功诱导其产生拓扑相变，并通过光谱定位器等手段揭示了局域原子配位对非晶拓扑材料的关键作用，从而确立了非晶碳作为工程化拓扑相的 versatile 平台。

Arnob Kumar Ghosh, Quentin Marsal, Annica M. Black-Schaffer2026-03-06  cond-mat.mes-hall

Altermagnets Enable Gate-Switchable Helical and Chiral Topological Transport with Spin–Valley–Momentum-Locked Dual Protection

Xianzhang Chen,^{1,2} Jiayong Zhang,^{1,2,3} Bowen Hao,¹ Jiahui Qian,¹

Ziye Zhu,^{1,2} Igor Žutić,⁴ Zhenyu Zhang,² and Tong Zhou^{1,*}

¹*Eastern Institute for Advanced Study, Eastern Institute of Technology, Ningbo, Zhejiang 3150200, P. R. China*

²*International Center for Quantum Design of Functional Materials (ICQD),*

and Hefei National Laboratory, University of Science and Technology of China, Hefei, 230026, China

³*School of Physical Science and Technology, Suzhou University of Science and Technology, Suzhou 215009, China*

⁴*Department of Physics, University at Buffalo, State University of New York, Buffalo, New York 14260, USA*
(Dated: February 20, 2026)

We establish a unified, symmetry-driven framework that combines the alternating spin splitting of altermagnets with valley topology to realize and electrically interconvert helical and chiral topological phases within a single material platform. We first demonstrate a magnetic analogue of the quantum spin Hall effect in altermagnets, hosting helical spin–valley–momentum–locked (SVML) edge states characterized by a composite spin–valley Chern number $C_{sv} = 2$. Large-scale quantum transport simulations show these SVML edge states exhibit fully quantized spin conductance robust against nonmagnetic and long-range magnetic disorder, reflecting their dual topological protection, while remaining vulnerable to short-range magnetic disorder. Exploiting that the counterpropagating SVML modes are linked by crystal rotation symmetry, we introduce a gate-tunable sublattice-staggered potential that selectively gaps one valley and converts the helical state into a chiral quantum anomalous Hall phase with $C_{sv} = 1$, robust against all disorder types. Reversing the potential switches the transmitted spin–valley polarization. Our first-principles calculations identify monolayer V_2STeO and VO families as realistic platforms supporting both helical and chiral topological phases and their electrical switching. These results establish altermagnets as electrically programmable platforms for robust topological devices across charge, spin, and valley.

The quantum spin Hall (QSH) effect hosts dissipationless helical edge states protected by time-reversal symmetry (TRS) [1, 2]. While spin–orbit coupling (SOC) is essential to open the topological gap, it simultaneously entangles spin channels and precludes strictly quantized spin Hall conductivity [1–4]. Moreover, even weak magnetic disorder breaks TRS, destabilizing helical edges and destroying quantized transport [5–8]. A common route to overcome this fragility is to convert QSH phases into quantum anomalous Hall (QAH) phases by introducing magnetism to break TRS [9–11]. However, this strategy typically requires long-range ferromagnetic order and finely tuned Chern gap reopening, severely constraining material realizations, operational temperatures, and device tunability [11, 12].

The emergence of altermagnets (AMs) offers a promising route to overcome these limitations [13–26]. AMs are collinear antiferromagnets with zero net magnetization yet spin-polarized bands, combining key advantages of ferromagnets and antiferromagnets [25–39]. Their momentum-space alternating spin polarization naturally separates spin channels by valley and suppresses inter-channel spin mixing, offering a rich landscape of robust spin–valley-related altermagnetic topological phases [40–55]. Unlike time-reversal-symmetric topological insulators, AMs intrinsically break TRS, opening new possibilities for realizing anomalous topological transport without invoking additional ferromagnetism [27, 38, 39]. Moreover, the relevant spin–valley sectors are connected by crystal rotation symmetries (CRS) [18, 25], suggesting a

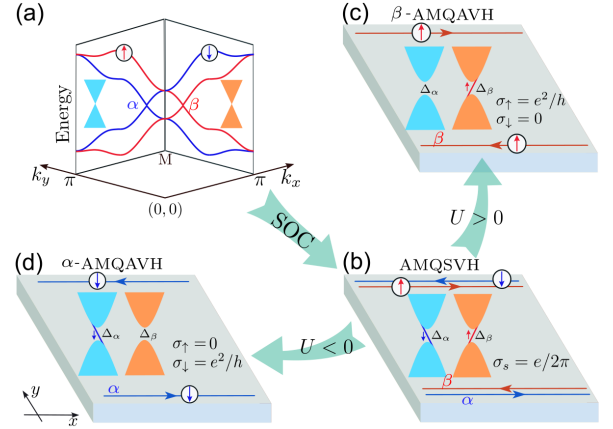


FIG. 1. (a) Schematic of spin–valley–resolved Weyl points at the α and β valleys in AMs. (b) SOC opens topological gaps at both valleys, yielding an AMQSVH phase with helical SVML edge states and fully quantized spin Hall conductivity ($\sigma_s = e/2\pi$). (c) A positive staggered potential ($U > 0$) selectively trivializes the gap at the α valley, eliminates its edge state, and produces a β -AMQAVH phase with spin-up chiral edge states at the β valley ($\sigma_\uparrow = e^2/h$). (d) Reversing the staggered potential ($U < 0$) yields an α -AMQAVH phase with spin-down chiral edge states at the α valley ($\sigma_\downarrow = e^2/h$).

symmetry-driven geometric handle to control topological phases and their transport behaviors. Despite these appealing features, a unified framework to describe and electrically manipulate distinct topological phases and their transport behaviors within a single AM platform re-

mainly lacking, particularly for rigorous topological characterization, robustness against disorder, and realistic material/device implementations.

Motivated by these opportunities, we develop a general framework that combines altermagnetic splitting with valley topology to realize and electrically switch helical QSH and chiral QAH states via selective valley gapping (Fig. 1). Starting from well-established AM Weyl semimetals [39, 53] with two valleys (α and β) [Fig. 1(a)], we show that SOC opens topological gaps at both valleys, yielding an altermagnetic QSH (AMQSVH) phase with spin-valley-momentum-locked (SVML) edge states as in Fig. 1(b). Unlike conventional QSH systems with only spin-momentum locking, SVML provides dual protection captured by a composite spin-valley Chern number $C_{sv} = 2$. Large-scale transport simulations confirm the AMQSVH phase keeps robust quantized conductance under nonmagnetic disorder and long-range magnetic fluctuations, reflecting its dual topological protection, but it is degraded by short-range (e.g. Anderson-type) magnetic disorder that can couple counterpropagating SVML channels.

Exploiting that the helical AMQSVH edge states are related by CRS, breaking this symmetry selectively gaps out one partner of the helical pair, thereby converting the AMQSVH into a chiral QAH phase without introducing ferromagnetism. We realize this transformation through a sublattice-staggered potential, U , which is natural in two-sublattice 2D materials and can be continuously controlled by an electric field [56–59]. This gate-tunable U enables selective valley gapping: by adjusting its strength, one valley is rendered topologically trivial while the other remains nontrivial, yielding a chiral SVML altermagnetic QAH (AMQAVH) phase with $C_{sv} = 1$, featuring gate-selectable spin-valley polarization, and robustness against all disorder types [Figs. 1(c)-(d)]. Guided by this symmetry-topology framework, our first-principles calculations further identify realistic monolayer families, including $V_2\text{STeO}$ and VO that support electrical switching between AMQSVH and AMQAVH phases. These results establish AM as a versatile platform for topological transport across charge, spin, and valley degrees of freedom.

We demonstrate our framework through an effective tight-binding model based on a 2D square AFM lattice as in Fig. 2(a), which is written as:

$$H = \sum_{i,j} \left[f_i^{\eta_j} c_i^\dagger c_{i+\eta_j} + g_i^{\kappa_j} c_i^\dagger c_{i+\kappa_j} + i\lambda_i^{\eta_j} c_i^\dagger \sigma_z c_{i+\eta_j} + h.c. \right] + M_{A,B} \sum_{i \in A,B} c_i^\dagger \sigma_z c_i + U_{A,B} \sum_i c_i^\dagger c_i. \quad (1)$$

Here, $c_i^\dagger$ (c_i) are electron creation (annihilation) operators at site i and σ is the Pauli matrix. The first and second terms describe hoppings between first (NN) and second (2NN) nearest-neighbors with hopping param-

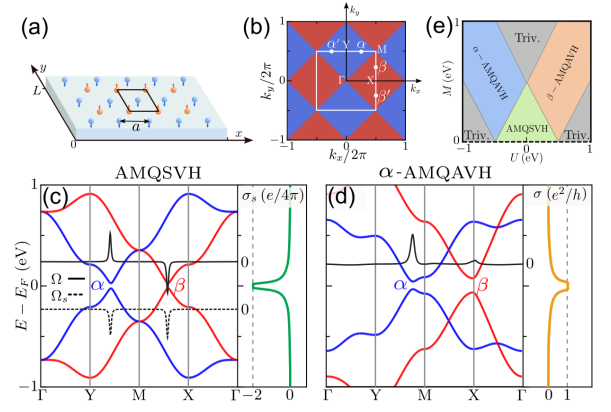


FIG. 2. (a) 2D square-lattice AM model for Eq. 1; the black square denotes the unit cell with lattice constant a . (b) Calculated d -wave AM spin texture in the first Brillouin zone, with valleys labeled α (α') and β (β'). (c) AMQSVH phase: spin-resolved bands (red/blue), Berry curvature Ω (black solid), spin Berry curvature Ω_s (black dashed), and the resulting quantized spin Hall conductivity σ_s . (d) Same as (c) but for the α -AMQAVH phase at $U < 0$, showing a quantized Hall conductivity σ . (e) Phase diagram as a function of U and M . Parameters are given in the main text and SM [60]

eters $f_i^{\eta_j}$ and $g_i^{\kappa_j}$, where η_j and κ_j are the vectors connecting site i to its NN and 2NN sites. The third term describes the SOC with strength $\lambda_i^{\eta_j}$. The fourth term indicates AFM exchange fields on sublattices A and B with $M_A = -M_B = M$; The last term, describes the staggered potential with $U_A = -U_B = U$. Unless stated otherwise, we take $M = 0.35$ eV and $|U| = 0.4$ eV throughout when these terms are included. All other parameters and model details are provided in the Supplemental Material (SM) [60].

Altermagnetism in this model is captured by sublattice-inequivalent hoppings together with the AFM exchange field [61–63], as confirmed by the calculated d -wave spin-polarization texture in Fig. 2(b). These terms also produce spin-valley-resolved Weyl points at the α (α') and β (β') valleys, consistent with previous studies [46, 54]. Here α (β) and α' (β') are related by mirror symmetry and exhibit identical spin splitting and topological character, as detailed in the SM [60]. For brevity, we henceforth refer to them collectively as the α and β valleys. These spin-valley-locked Weyl features, faithfully reproduced by our model, serve as the precursor to the AMQSVH and AMQAVH phases proposed below. Importantly, similar AM Weyl textures arise in realistic materials such as monolayers CrO [41, 42], Nb_2SeTeO [45], $\text{Fe}_2\text{Te}_2\text{O}$ [47], and their derivatives [53], underscoring the generality and material relevance of our theoretical framework.

When SOC is introduced, each Weyl point is gapped as shown in Fig. 2(d). The resulting Berry curvature $\Omega(\mathbf{k})$ becomes strongly concentrated near α and

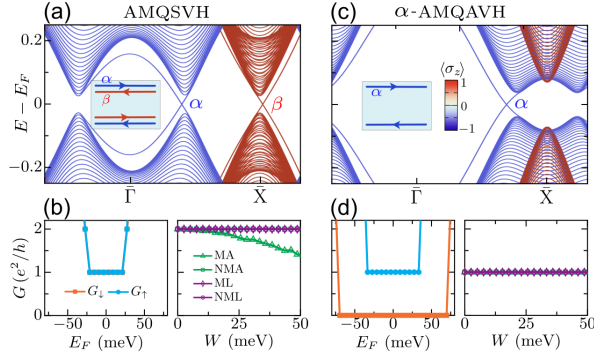


FIG. 3. (a) Nanoribbon bands of the AMQSVH phase, showing helical SVML edge states located at α and β valleys. (b) Spin-resolved conductance for (a) as a function of Fermi energy E_F and disorder strength W , comparing magnetic Anderson (MA), nonmagnetic Anderson (NMA), magnetic long-range (ML), and nonmagnetic long-range (NML) disorder. (c,d) Same as (a,b) but for the α -AMQAVH phase with a single chiral edge channel. Parameters taken from Fig. 2.

β valleys with opposite signs, yielding a vanishing total Chern number but a quantized valley Chern number $C_v = C_\alpha - C_\beta = 2$ [59, 64], where C_α (C_β) is the local topological charge around α (β) valley. Because spin-up (spin-down) bands are confined to the α (β) valley, spin mixing is strongly suppressed allowing spin to remain a good quantum number. The corresponding spin Berry curvature $\Omega_s(\mathbf{k})$ integrates to a fully quantized spin Hall conductivity $\sigma_s = -e/2\pi$, characterized by a spin Chern number [65–68], $C_s = C^\uparrow - C^\downarrow = -2$. The coexistence of nonzero C_s and C_v reveals a new class of spin-valley-resolved QSH states endowed with dual topological protection.

To elucidate the topology of this phase, we examine the low-energy form of Eq. (1). Expanding the Hamiltonian near each valley yields a continuum model

$$H = \hbar v_F (k_x \tau_x + k_y \tau_y) + \gamma \lambda \tau_z, \quad (2)$$

where v_F is the Fermi velocity, $\gamma = \pm 1$ labels the spin-valley sector, λ is the SOC strength, and $\boldsymbol{\tau}$ denotes Pauli matrices acting on the conduction and valence bands. Integrating the spin-valley-resolved $\Omega(\mathbf{k})$ around each spin-valley pocket gives topological charges $(C_\alpha^\downarrow, C_\beta^\uparrow) = (1, -1)$, from which $C_s = C_v = 2$. Since either index alone cannot fully capture the joint topology, we define a composite spin-valley Chern number

$$C_{sv} = C_\alpha^\downarrow - C_\beta^\uparrow, \quad (3)$$

which uniquely characterizes the spin-valley-locked band topology. For this system $C_{sv} = 2$, implying a pair of helical SVML edge states. The calculated nanoribbon spectra in Fig. 3(a) and wave function probabilities in SM [60] show that spin-up (spin-down) edge states are indeed bound to the α (β) valley, directly confirming

the helical SVML AMQSVH states. This dual locking suppresses spin mixing and yields a genuine realization of the long-sought ideal QSH effect, verified by transport simulations showing robust quantized spin Hall conductance [Fig. 3(c)]. Unlike conventional QSH or QVH systems, which rely solely on spin- or valley-momentum locking [1, 59], the AMQSVH phase combines both and can offer superior robustness against disorder.

Beyond the dual SVML protection, altermagnets provide further key advantage: the two counterpropagating SVML edge branches are linked by CRS [18, 39]. As a result, breaking this symmetry can gap out one branch and convert the helical AMQSVH phase into a chiral AMQAVH phase, without introducing ferromagnetism or relying on finely tuned Chern-gap reopening as in conventional schemes [9–11]. We realize this symmetry-controlled transition through U , a generic rotation-symmetry-breaking perturbation widely used for band-structure engineering and topological phase manipulation in two-sublattice 2D materials, and readily tunable by an external electric field [8, 56–59]. The U -dependent band structures in [Fig. 2(d)] and the phase diagram in [Fig. 2(e)] (see more details in SM [58]) show that for $U < 0$ the β -valley gap closes and reopens in a trivial form, while the α -valley gap remains nontrivial, yielding an α -AMQAVH phase as corroborated by the Berry-curvature distribution and the quantized Hall conductance in Fig. 2(d). Correspondingly, the helical edge pair collapses into a single chiral SVML channel with fixed spin–valley polarization (spin down in the α valley), producing disorder-robust chiral transport [Fig. 3(c-d)]. Reversing the sign of U instead selects the β -AMQAVH phase with the opposite spin–valley polarization (see SM [60]). Because both the magnitude and sign of U can be continuously tuned by gating [8, 56–59], these results realize an all-electrical switch between helical and chiral topological transport, positioning altermagnets as a scalable platform for electrically reconfigurable spin–valley topological devices.

A defining hallmark of topological phases is their disorder-resilient transport. The robustness of quantized conductance under specific types of disorder provides both a fingerprint of the underlying topology and a criterion for device utility [1, 2, 59]. For instance, conventional QSH states protected by TRS remain dissipationless under nonmagnetic disorder but are destroyed by magnetic impurities that break TRS and disrupt spin–momentum locking [1, 2]. Similarly, quantum valley Hall states are robust against long-range disorder yet fragile under short-range disorder that induces intervalley scattering and lifts valley–momentum locking [8, 59]. To evaluate the disorder robustness of the AMQSVH state, we calculate spin-resolved conductance using the Landauer–Büttiker formalism and Green’s-function method [69, 70]. We consider both magnetic and nonmagnetic disorder—distinguished by their abil-

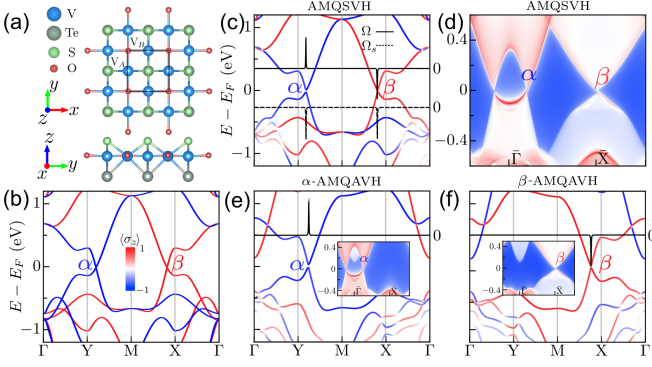


FIG. 4. (a) Top and side views of monolayer V_2TeO . (b) Spin-resolved bands of (a) without SOC, showing AM Weyl points. (c) Same as (b) but with SOC; the corresponding (spin) Berry curvatures indicate the AMQSVH phase, whose helical edge states are shown in (d). (e) Same as (c) but for Ti substitution on the V_A sublattice, which induces $U < 0$ and yields the α -AMQAVH phase with chiral edge state (inset). (f) Same as (e) but for Ti substitution on the V_B sublattice ($U > 0$), yielding the β -AMQAVH state.

ity to flip spins—and short- versus long-range correlation lengths, ζ , relative to the lattice constant a . Conductance calculations for AMQSVH nanoribbons under four representative disorder types (magnetic/nonmagnetic Anderson, $\zeta=0$; magnetic/nonmagnetic long-range, $\zeta=5a$) are shown in Fig. 3(c), and more details are given in the SM [60]. Consistent with its dual protection, the AMQSVH phase retains quantized conductance under both short- and long-range nonmagnetic disorder, similar to conventional QSH systems. However, unlike QSH phases that are easily destroyed by long-range magnetic fluctuations [2, 71], AMQSVH remains quantized because its additional valley-momentum locking protection survives. Only magnetic Anderson disorder—capable of simultaneous spin flips and intervalley scattering—breaks the SVML condition and suppresses quantization. These results directly confirm that dual SVML protection is responsible for the enhanced robustness of the AMQSVH state. To overcome this remaining fragility against magnetic Anderson disorder, one can selectively gap a single valley to remove one helical branch, driving the system into the chiral AMQAVH phase discussed above. Unlike conventional QAH states where spin and valley indices are ill-defined, AMQAVH preserves SVML character and exhibits quantized spin-valley-polarized chiral transport robust against all disorder types [Fig. 3(d)].

Experimentally, our proposed AMQSVH and AMQAVH phases, together with their electrically controlled interconversion, are well supported by realistic material platforms. Guided by the symmetry criteria and model insights developed above, our first-principles calculations identify monolayers V_2STeO , VO , and related derivatives as AMQSVH candidates (see SM [60]). As a representative example, we consider

monolayer V_2STeO to illustrate a concrete realization: without SOC its spectrum hosts spin-valley-locked Weyl points [Fig. 4(b)], whereas SOC opens gaps of order 20 meV at both valleys [Fig. 4(c)], producing the targeted AMQSVH phase. The corresponding (spin) Berry-curvature distributions and Wannier-based edge-state calculations, which reveal helical SVML edge modes [Figs. 4(c,d)], further confirm this identification.

To realize the chiral AMQAVH state, we introduce a controllable sublattice asymmetry that generates an effective U . Concretely, we implement this via alloy engineering, replacing the V atom on one sublattice (V_A) by Ti , a standard strategy for engineering sublattice potentials and tuning band topology [8, 56–59]. This substitution selectively trivializes the β -valley gap while preserving a topological gap at α valley, yielding the α -AMQAVH phase as confirmed by the Berry curvature and the emergence of a single chiral SVML edge channel [Fig. 4(e)]. Replacing instead the V atom on the opposite sublattice (V_B) reverses the effective sign of U and switches the system to the β -AMQAVH phase [Fig. 4(f)], directly validating the symmetry-based design principle. This material route to AMQAVH state is not specific to V_2STeO and can extend to other monolayer families (e.g. Nb_2SeTeO and VO) as seen in SM [60], underscoring the universality and experimental feasibility of our proposal.

By integrating three defining ingredients of altermagnets—valley-locked spin splitting, rotation-symmetry-connected valleys, and intrinsic antiferromagnetic order—we establish a unified and symmetry-driven framework for realizing and electrically interconverting helical and chiral topological transport within a single material platform. The resulting SVML edge channels provide dual topological protection, yielding enhanced robustness against disorder while enabling deterministic, electrical selection of the transmitted channel type and its spin-valley polarization. More broadly, because our proposed mechanism is rooted in symmetry and band topology rather than material-specific chemistry, it is readily extendable to topological photonic [72], phononic [73], magnonic [74], and superconducting systems [75–77]. This scalability opens a pathway toward low-dissipation, electrically programmable topological circuitry and multifunctional quantum devices [78], bridging condensed-matter physics, materials science, and emerging quantum technologies.

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X.C., J.Z., and B.H. contributed equally to this work.

* tzhou@eitech.edu.cn

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